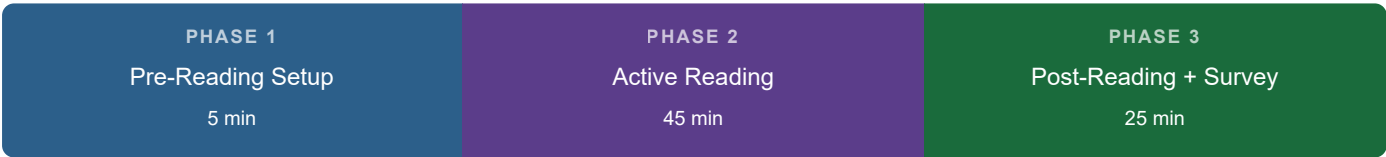


What We Are Finding from the *User Study*

A complete data collection and outcome framework for the CoRead evaluation.
Designed to meet UIST best-paper standards for systems evaluation.

PARTICIPANTS	CONDITIONS	SESSIONS
12 (within-subjects)	CoRead vs. Baseline	24 total (2 per participant)
SESSION LENGTH	METHOD	ANALYSIS
75 minutes each	WoZ confederate + solo	Wilcoxon + Spearman + Thematic

1. Study Design Overview



Each of the 12 participants completes **two sessions** (one CoRead, one Baseline) separated by at least one week. Order is counterbalanced: 6 participants do CoRead first, 6 do Baseline first. Two papers of matched complexity are used, also counterbalanced.

CoRead sessions use a **Wizard-of-Oz confederate protocol**: the researcher operates two additional authenticated accounts in separate browser windows, simulating peer collaborators following a standardized behavioral script. This ensures every participant experiences an identical collaborative context — validated practice in groupware research (Gutwin & Greenberg, 1998).

WHY WITHIN-SUBJECTS WITH N=12?

Within-subjects design controls individual differences and doubles statistical power at small N. Wilcoxon Signed-Rank tests are appropriate (non-parametric, paired, N=12). Standard at UIST/CHI for lab-based systems evaluation.

2. Data Sources

Source	What it captures	Storage	Both conditions?
System logs	Every user interaction, timestamped (studyLogger.ts)	localStorage + /study-data/ JSON	CoRead only
Existing activity tracker	Page visits, AI calls, scroll, highlight events	localStorage (activityTracker.ts)	CoRead only
Post-session survey	NASA-TLX, perceived quality, AI timing ratings	Google Form / paper form	Both conditions
Screen recording	Full session video — for behavioral coding	OBS / Zoom local recording	Both conditions
Post-study interview	Qualitative: perceptions, preferences, incidents	Audio recording + transcript	Once, after both sessions

3. Category A — Feature Usage & Adoption

The primary question: *did participants actually engage with CoRead's novel features, and which ones?* A reviewer's biggest concern for systems papers is that participants ignore AI features and use the tool like a basic PDF reader. This data directly refutes or confirms that concern.

A1. Feature Adoption Table

Reported as: **N out of 12 participants who used each feature at least once**, plus mean uses per session ($M \pm SD$).

Feature	Category	N / 12	M uses/session	SD	First use (min)
Role Assignment — viewed	Role	—	—	—	—
Role Assignment — accepted	Role	—	—	—	—
Role Assignment — modified	Role	—	—	—	—
Highlight — any type	Annotation	—	—	—	—
Highlight — Confused	Annotation	—	—	—	—
Highlight — Important	Annotation	—	—	—	—
Highlight — Understood	Annotation	—	—	—	—
Highlight — Question	Annotation	—	—	—	—
Stuck Marker	Annotation	—	—	—	—
AI Help (general)	AI	—	—	—	—
AI Math Explainer	AI	—	—	—	—
AI Image / Figure Analysis	AI	—	—	—	—
AI Tutor	AI	—	—	—	—
AI Summary	AI	—	—	—	—
Share Panel opened	Collaboration	—	—	—	—
Notification clicked	Notification	—	—	—	—
Peer Chat accepted	Collaboration	—	—	—	—
Discussion Report opened	Post-reading	—	—	—	—
Team Consensus opened	Post-reading	—	—	—	—
Collective Knowledge opened	Post-reading	—	—	—	—

A2. Feature Usage Trajectory

Line chart showing mean usage frequency per 10-minute block across the 45-minute session. Reveals *when* each feature is most used — e.g., AI Help peaks in minutes 20–30 (dense methodology sections), Post-reading artifacts peak at session end. Confirms features are used at the right moments.

A3. Time-to-First-Use (Discoverability)

Horizontal bar chart: one bar per feature, showing median minutes into session before first use. Features discovered within the first 5 minutes = highly discoverable. Features first used after 30 minutes = potentially hidden or intimidating.

A4. Feature Abandonment

For each feature: what % of participants who used it once used it a second time? Low return rate (e.g., 1/10 returned) = feature was tried but not found useful. High return rate = genuine ongoing utility.

4. Category B — AI Timing Accuracy

This is CoRead's most **novel and publishable metric**. No prior paper has measured whether AI assistance arrived at the right moment in a reading session. Three complementary methods triangulate the answer.

METHOD 1

Log-based

Time from CIM trigger to participant's next action. Short gap = timely.

METHOD 2

Behavioral

Notification accepted vs dismissed.
Acceptance = well-timed signal.

METHOD 3

Retrospective

Participant rates each notification after session: well-timed / too early / too late.

B1. Notification Outcome Breakdown

Outcome	Definition	N events	%
Clicked (engaged)	Participant opened notification within 30s	—	—
Read then dismissed	Notification visible >2s then closed	—	—
Ignored (fast dismiss)	Dismissed within 2s — not read	—	—
Auto-expired	Auto-dismissed after 6s without interaction	—	—

B2. Timing Accuracy by Struggle Severity

Compare acceptance rate when CIM fired at medium severity (score 0.72–0.88) vs. high severity (score >0.88). If high severity → higher acceptance rate, this validates the CIM threshold calibration.

B3. Retrospective Timing Ratings

After the session, each participant sees a timeline of every notification that fired during their session and rates each: *Well-timed* / *Too early* / *Too late* / *I didn't notice it*. Reported as percentage distribution across all ratings from all 12 participants.

WHY THIS MATTERS TO REVIEWERS

Prior work on AI assistance timing (Horvitz 1999) identified timing as the central challenge of mixed-initiative systems but provided no empirical measurement method. CoRead's CIM gives a formal trigger; this data is the first empirical test of whether that trigger actually fires at the right moment.

5. Category C — User Control & Opt-Out Behaviors

Measures whether participants exercised control over CoRead's monitoring and collaboration features — and whether they felt comfortable with the system's level of access. Directly addresses reviewer questions about privacy and surveillance concerns in behavioral AI systems.

Feature	Opt-out event	N opted out	N opted back in	Net opt-out	Insight
Implicit AI help	<code>implicit_ai_disabled</code>	—	—	—	Proactive help tolerance
Struggle broadcasting	<code>struggle_broadcast_off</code>	—	—	—	Confusion privacy concern
Collaboration mode	<code>collaboration_mode_off</code>	—	—	—	Preference for solo reading
Annotation broadcasting	<code>annotation_broadcast_off</code>	—	—	—	Sharing highlights comfort
Eye tracking	<code>eye_tracking_declined</code>	—	—	—	Gaze monitoring comfort
Awareness panel	<code>awareness_panel_hidden</code>	—	—	—	Others' highlights distraction
Notifications	<code>notifications_muted</code>	—	—	—	Notification fatigue

THE "OPTED BACK IN" COLUMN IS YOUR MOST INTERESTING FINDING

A participant who turns a feature off and then re-enables it demonstrates experiential trust-building — they needed to see the system behave before trusting it with their data. This is a design insight no survey question can capture.

Privacy Sensitivity Hierarchy

Rank features by opt-out rate. Expected finding based on formative study: *struggle broadcasting* will have the highest opt-out (participants don't want to broadcast confusion) while *annotation broadcasting* will have lower opt-out (sharing highlights feels safer). This validates the formative finding that confusion signals carry higher social cost than content signals.

6. Category D — Behavioral Outcomes

Quantitative measures derived from recordings and logs, compared CoRead vs. Baseline using Wilcoxon Signed-Rank tests.

Measure	Definition	Source	CoRead M (SD)	Baseline M (SD)	Expected direction
Confusion highlights	Highlights tagged as "Confused" per session	Log	—	—	CoRead lower (resolved faster)
Total highlights	All annotation events per session	Log	—	—	CoRead higher (richer engagement)
Section coverage	Unique sections annotated / total sections	Log	—	—	CoRead higher (role assignment works)
Assigned section depth	Highlights in assigned section vs. other sections	Log	—	—	CoRead: assigned > other
Time-to-first-annotation	Min until first substantive highlight	Log	—	—	CoRead shorter (roles reduce cold start)
AI help invocations	Explicit AI requests per session	Log	—	N/A	—
Post-summary dwell	Time spent on Discussion Report artifacts	Log	—	N/A	—
Revisit rate	% of pages visited more than once	Log	—	—	CoRead lower (less rereading needed)

7. Category E — Subjective Measures (Surveys)

E1. NASA-TLX — both conditions

Six subscales (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, Frustration), each 0–100. Administered after each condition. Key hypothesis: CoRead reduces Mental Demand and Frustration without increasing Temporal Demand.

E2. Perceived Collaborative Quality — both conditions

Custom 5-point Likert scale (5 items):

- "Our group worked well together during this session"
- "I felt comfortable asking for help when confused"
- "I understood what my teammates were focusing on"
- "Our discussion covered the paper thoroughly"
- "I felt my contributions were valuable to the group"

E3. AI Feature Ratings — CoRead condition only

Item	Scale	Constructs
"The AI assistance appeared at the right moment"	1–5	AI timing accuracy
"The AI notifications were distracting"	1–5 (R)	Notification intrusiveness
"The AI help was relevant to what I was confused about"	1–5	AI relevance
"I felt the AI understood when I needed help"	1–5	AI empathy / appropriateness
"I would have wanted more AI help than I received"	1–5	Under-intervention sensitivity
"I would have wanted less AI help than I received"	1–5	Over-intervention sensitivity
"The role assignment helped me focus my reading"	1–5	Role assignment utility
"The post-reading summary helped start our discussion"	1–5	Synthesis artifact utility
"The peer chat invitation came at a useful moment"	1–5	Peer brokering timing

E4. Post-Study Preference (once, after both conditions)

- "Which condition did you prefer overall?" (CoRead / Baseline / No preference)
- "Which helped you understand the paper better?" (same options)
- "Which would you use in your real reading group?" (same options)

8. Category F — Individual Differences & User Archetypes

With 12 participants you can identify **usage archetypes** — qualitatively distinct patterns of engagement that reveal how different users appropriate the system.

Archetype	Signature pattern	Design implication
Heavy AI users	≥5 AI invocations, long response dwell, math explainer used	AI features support diverse expertise levels
Annotation-first users	Many highlights, few AI invocations, Share Panel used heavily	Collaborative annotation valued over AI assistance
Passive readers	Few highlights, few AI calls, mostly page-visit events	System may need stronger onboarding nudges
Trust-builders	Opted out of features early, re-enabled after 10–15 min	Experiential trust more effective than upfront explanation

WHY ARCHETYPES MATTER

Showing that CoRead accommodated multiple reading styles — rather than forcing a single interaction model — is a strong design argument. It addresses the reviewer concern: "What about participants who don't want AI help?" The archetype data shows those participants can use CoRead successfully in annotation-first mode.

9. Category G — Correlation Analysis

Spearman rank correlations (non-parametric, appropriate for N=12) between feature usage and outcomes. These answer the question: *did using certain features actually lead to better results?*

Variable 1	Variable 2	Hypothesis
AI Help invocations	NASA-TLX Mental Demand	More AI help → lower cognitive load
Confusion highlights	Subsequent AI help in same section	Confusion annotation predicts help-seeking
Role Assignment accepted	Annotation depth in assigned section	Role ownership → deeper engagement with assigned content
Notification clicked	AI timing accuracy survey rating	Behavioral and subjective timing measures should align
Struggle score at CIM trigger	Notification acceptance rate	Higher struggle → more timely intervention (validates CIM)
Post-summary dwell time	Perceived Collaborative Quality	Engaging with artifacts → better perceived collaboration
Opt-out count	Overall CoRead preference	Users who exercised more control feel more positive

10. Category H — Qualitative Findings

Semi-structured interview (15 min) conducted once after both sessions. Analyzed using reflexive thematic analysis (Braun & Clarke) with two coders. Target: 4–6 themes with representative quotes.

Interview Questions

- "Walk me through a moment in the CoRead session where the AI did something that felt helpful or unhelpful."
- "Did the role assignment change how you read compared to the other session?"
- "Was there a moment where you wished AI had stepped in but it didn't? Or stepped in when you didn't want it to?"
- "How did the post-reading summary affect your group discussion?"
- "You turned off [X feature] during the session — can you tell me why?" (*only if they opted out of something*)
- "Would you use CoRead in your actual reading group? What would you change?"

Expected Themes (grounded in formative study)

- **Epistemic accountability** — role assignment creates responsibility to read deeply
- **Trust calibration** — comfort with AI monitoring develops during the session, not before
- **Confusion visibility** — seeing confusion signals reduces social cost of asking for help
- **Artifact-anchored discussion** — post-reading synthesis changes how discussion starts
- **Attention economy** — relevance-filtered notifications preserved focus better than expected

11. Paper Outputs — Tables & Figures

#	Output	Type	Data source
T1	Feature adoption & usage statistics (all features, $M \pm SD$, $N/12$)	Table	System logs
T2	CoRead vs. Baseline quantitative comparison (W , p , r for each measure)	Table	Logs + survey
T3	AI feature subjective ratings ($M \pm SD$ for each item)	Table	Survey
T4	User control & opt-out behaviors (N opted out, N re-enabled)	Table	System logs
T5	Spearman correlations: feature usage vs. outcomes	Table	Logs + survey
F1	Feature adoption bar chart (sorted by % adoption)	Figure	System logs
F2	Feature usage trajectory (usage/10min across session)	Figure	System logs
F3	AI timing accuracy breakdown (well-timed / too early / too late / ignored)	Figure	Logs + survey
F4	Time-to-first-use per feature (discoverability chart)	Figure	System logs
F5	NASA-TLX subscale comparison CoRead vs. Baseline	Figure	Survey

12. Statistical Analysis Plan

Comparison	Test	Report
CoRead vs. Baseline on any continuous measure	Wilcoxon Signed-Rank (within-subjects, $N=12$)	W , p , $r = Z/\sqrt{N}$
Feature usage vs. outcome variable	Spearman rank correlation	ρ , p , N
Notification outcome distribution	Descriptive ($\%$, counts)	Table + figure
Multiple comparisons (≥ 3 tests on same data)	Bonferroni correction	Adjusted p reported
Qualitative coding agreement	Cohen's κ	Target $\kappa > 0.75$

IMPORTANT FOR SMALL N

With $N=12$, report all tests as exploratory rather than confirmatory. State: *"Given the small sample size, we report Wilcoxon tests and effect sizes as exploratory indicators; findings should be interpreted with appropriate caution and validated in larger deployments."* Reviewers expect this honesty and will penalize papers that over-claim statistical significance at $N=12$.